



Module 2

Multi-Label Classification



Multi-Label Classification



Types of Classification Problems

- The 3 main classification problems are:

Binary
Classification



- Spam
- Not spam

Multiclass
Classification



- Dog
- Cat
- Horse
- Fish
- Bird
- ...

Multi-label
Classification



- Dog
- Cat
- Horse
- Fish
- Bird
- ...

Multi-Label Classification

- is the supervised learning problem where an instance may be associated with multiple labels

Sample	Class	Sample	Classes
	Red		Red, Blue, Yellow
	Green		Yellow, Green
	Blue		Blue, Pink, Yellow

a

b

Multi-Label Classification

Feature Matrix (X)

n_features →

← n_samples

← n_samples

0
0
1
0
1
0
1
1
0

← n_samples

1
0
0
1
0
1
0
1
0

← n_samples

1
1
0
0
1
0
0
1
0



Response matrix

Binary Relevance

- ❑ Decomposes learning tasks into independent binary problems.
- ❑ Returns a propensity/prediction vector for each response
- ❑ We fit each response independently

X	Y_1	Y_2	Y_3	Y_4
$X^{(1)}$	0	0	0	1
$X^{(2)}$	1	0	0	0
$X^{(3)}$	1	0	0	1
$X^{(4)}$	0	1	0	0
$X^{(5)}$	0	1	1	0

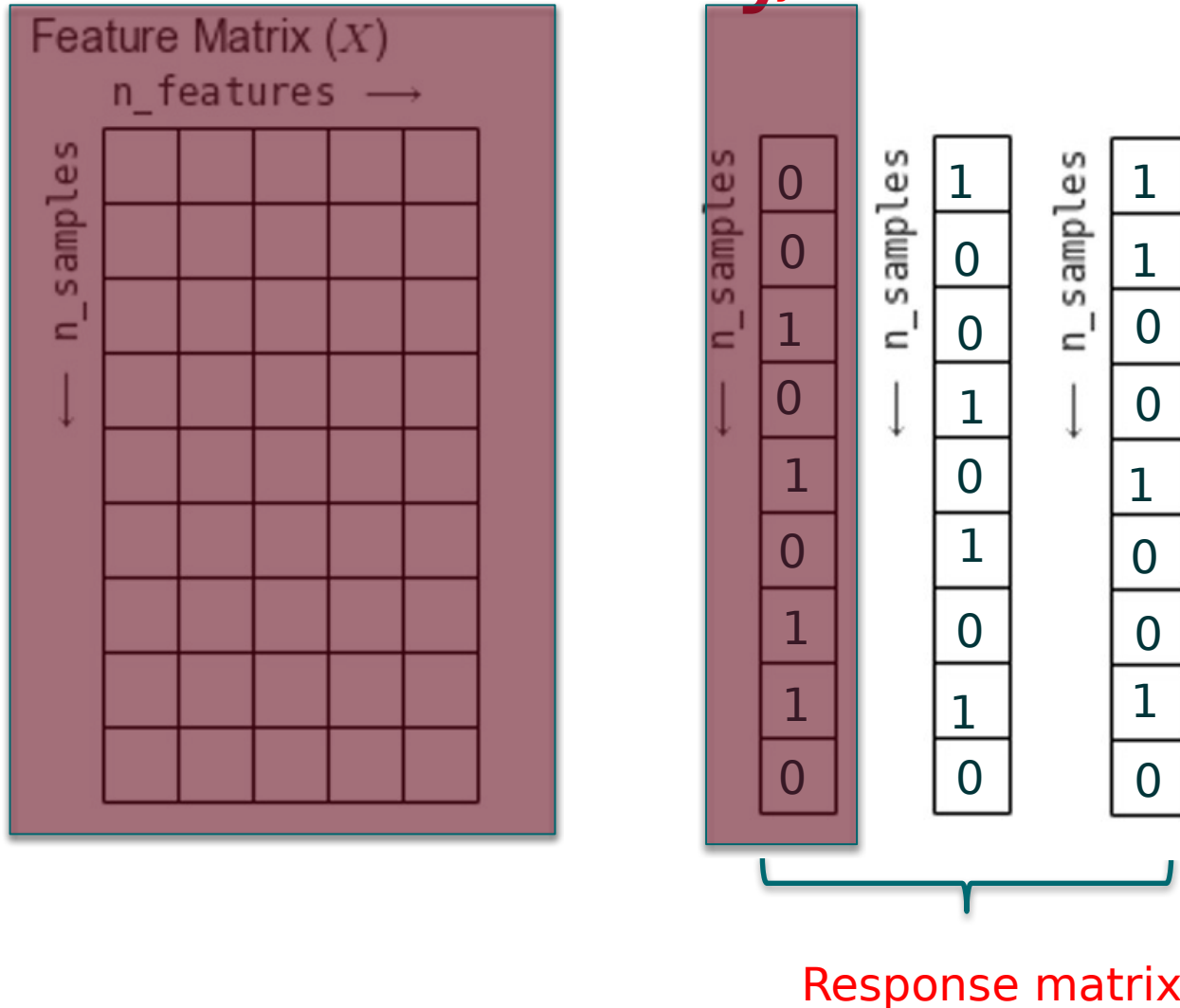


X	Y_1
$X^{(1)}$	0
$X^{(2)}$	1
$X^{(3)}$	1
$X^{(4)}$	0
$X^{(5)}$	0

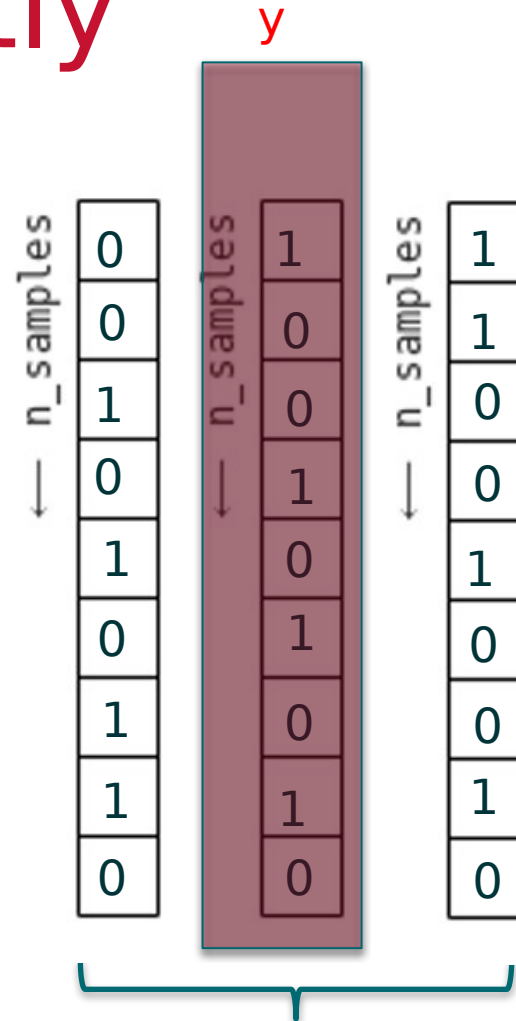
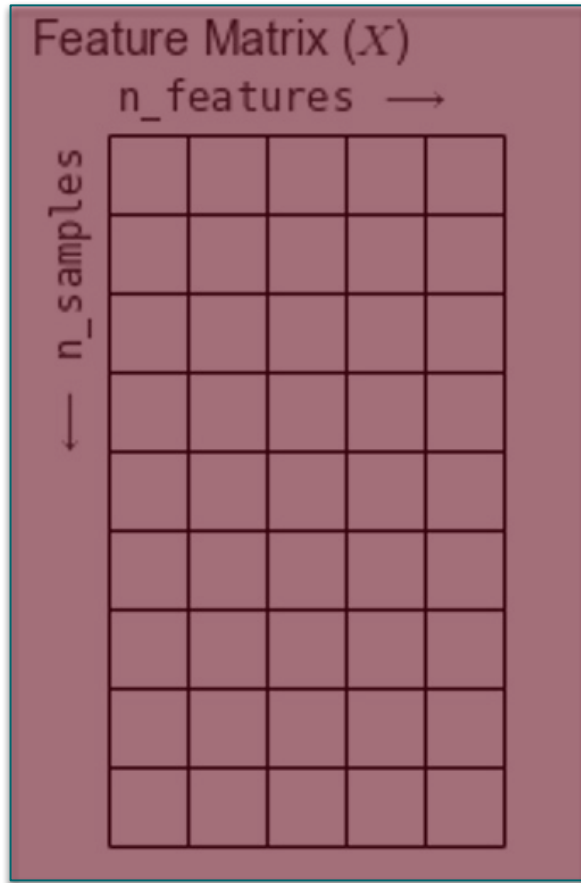
X	Y_2
$X^{(1)}$	0
$X^{(2)}$	0
$X^{(3)}$	0
$X^{(4)}$	1
$X^{(5)}$	1

X	Y_3
$X^{(1)}$	0
$X^{(2)}$	0
$X^{(3)}$	0
$X^{(4)}$	0
$X^{(5)}$	1

Fitting Each Response Independently

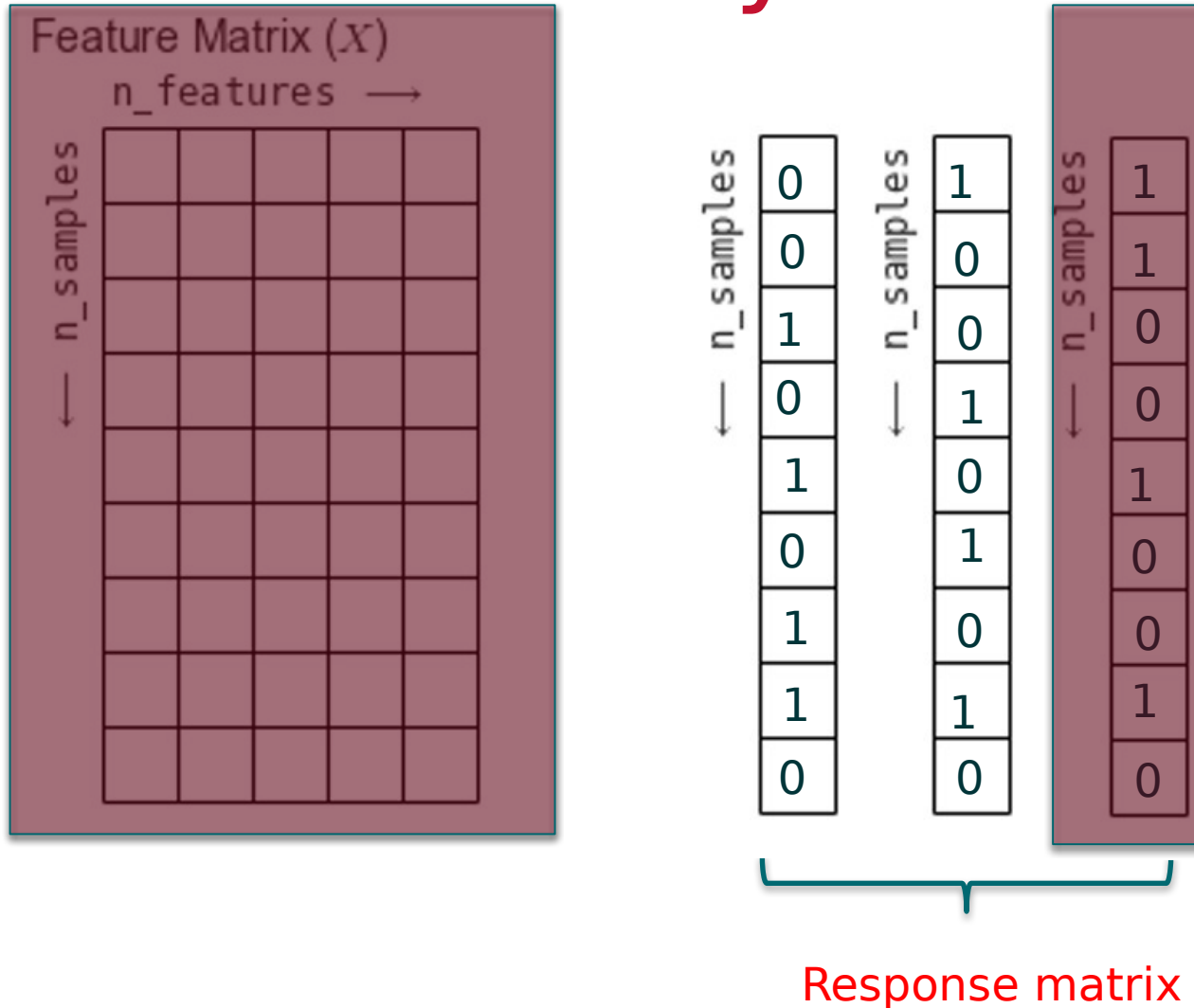


Fitting Each Response Independently



Response matrix

Fitting Each Response Independently





Python

Classifier Chains

- Arrange binary classifiers into a chain adding response vector to features.

X	Y_1	Y_2	Y_3	Y_4
X_1	0	1	0	0
X_2	0	1	1	0
X_3	1	0	0	0
X_4	0	1	0	0



X	Y
X_1	0
X_2	0
X_3	1
X_4	0

X	Y_1	Y_2
X_1	0	1
X_2	0	1
X_3	1	0
X_4	0	1

X	Y_1	Y_2	Y_3
X_1	0	1	0
X_2	0	1	1
X_3	1	0	0
X_4	0	1	0

X	Y_1	Y_2	Y_3	Y_4
X_1	0	1	0	0
X_2	0	1	1	0
X_3	1	0	0	0
X_4	0	1	0	0

Multi-Label Classification

Feature Matrix (X)

$n_features \rightarrow$

$\leftarrow n_samples$

$\leftarrow n_samples$

0
0
1
0
1
0
1
0
1
1
0

$\leftarrow n_samples$

1
0
0
1
0
1
0
1
0
1
0

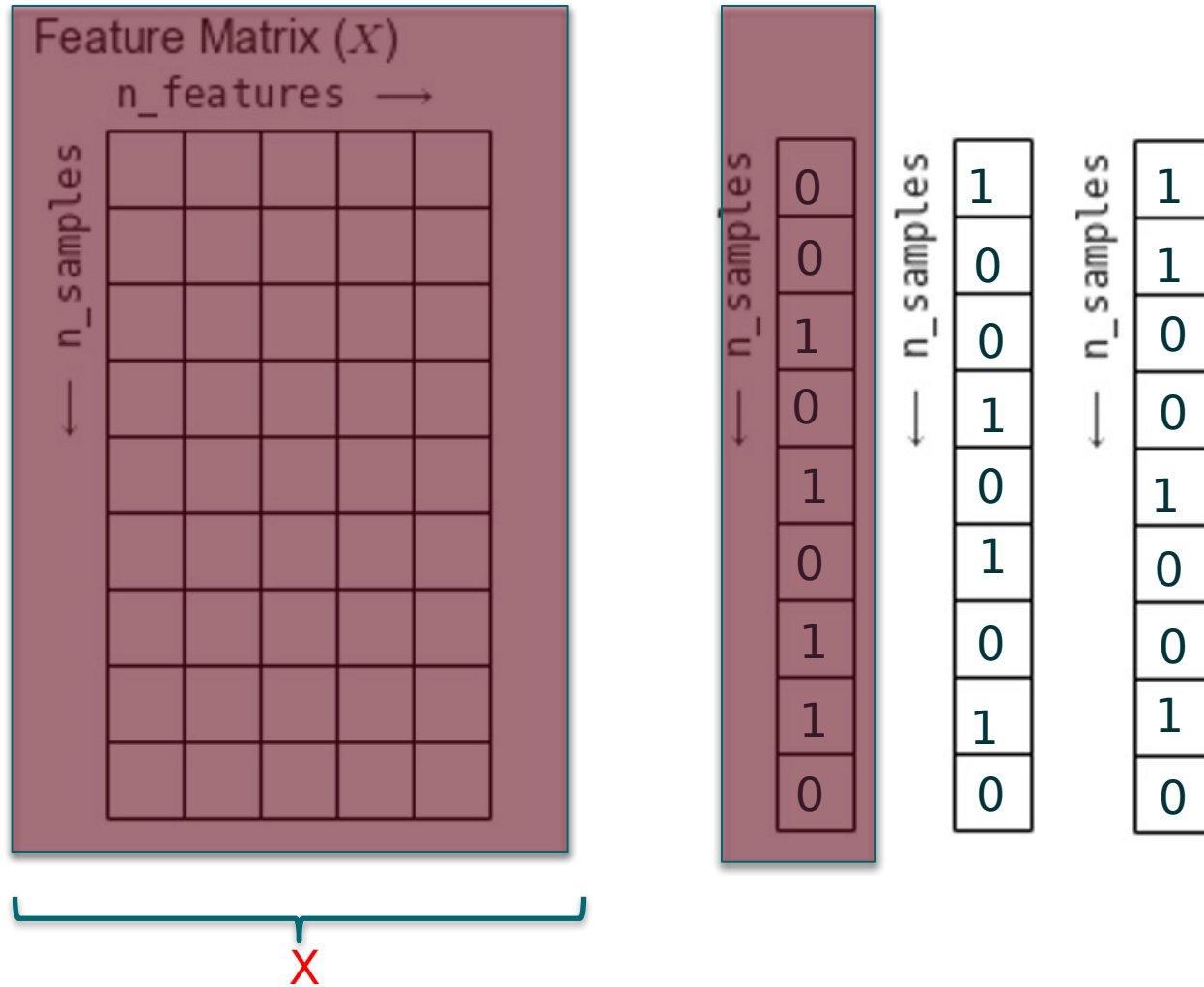
$\leftarrow n_samples$

1
1
0
0
1
0
0
1
0
1
0

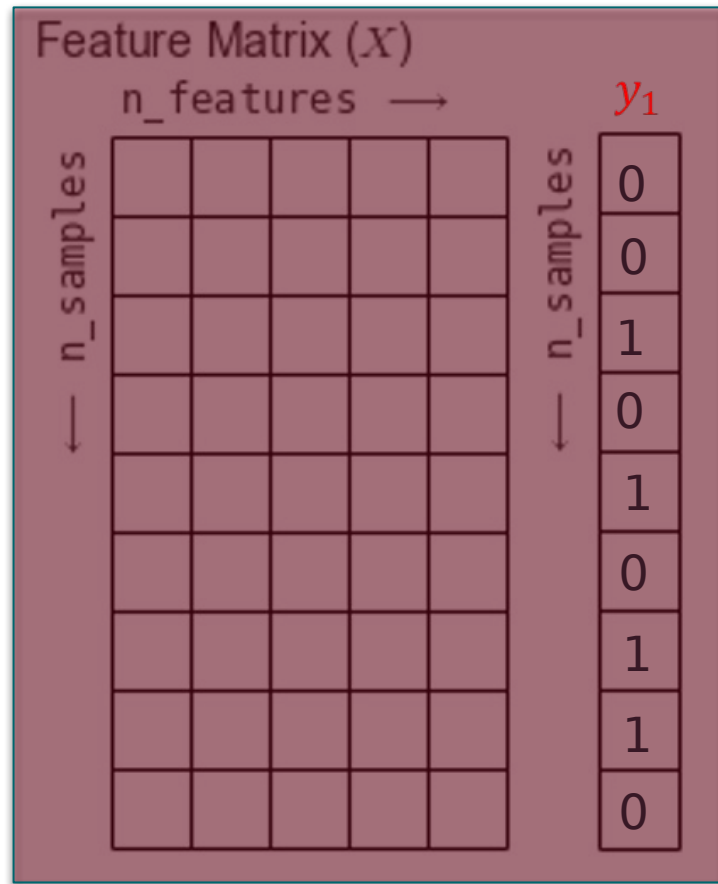


Response matrix

Creating a Chain

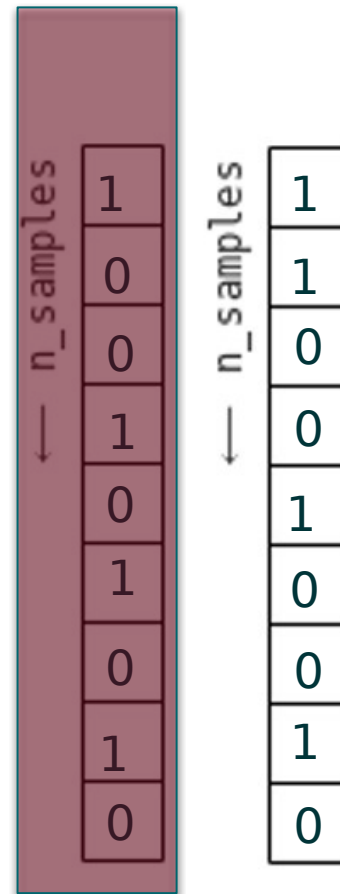


Creating a Chain



X

y





Python

Metrics: Exact Match Ratio

- ❑ Predictions that are exact matches to the response vectors are considered accurate
- ❑ Partial matches are considered errors

[illegible]



Python

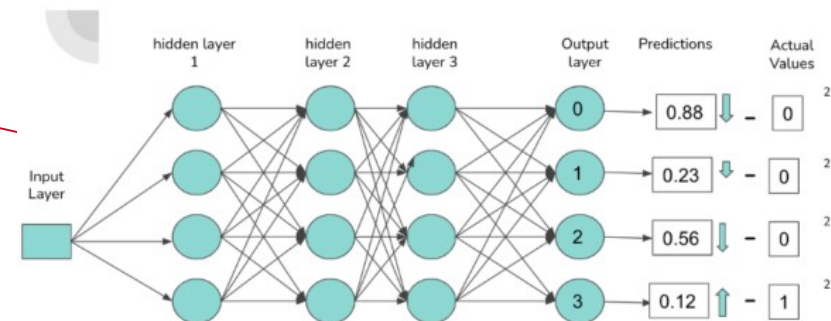
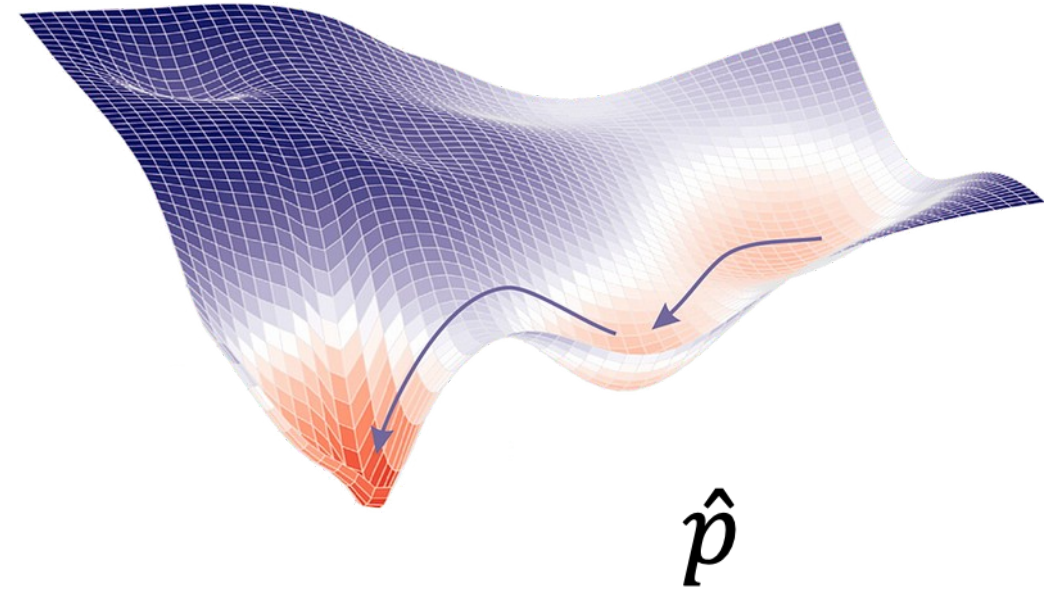
Common Cost Functions Neural Nets

- Binary Cross-Entropy
- Loss Function

$$J(w) = -\frac{1}{n} \sum (y \log \hat{p} + (1 - y) \log(1 - \hat{p}))$$

- Gradient

$$\nabla J(w) = \hat{p} - y$$



Classify the Number "3"

Activation Functions Neural Nets

- Common Activations on Layers

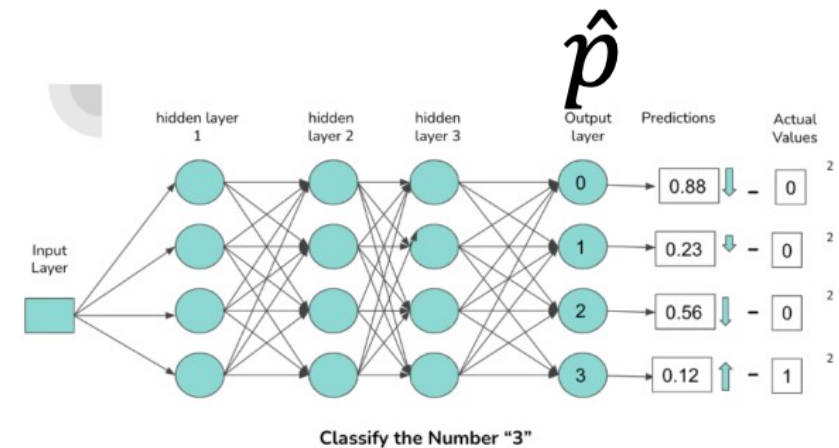
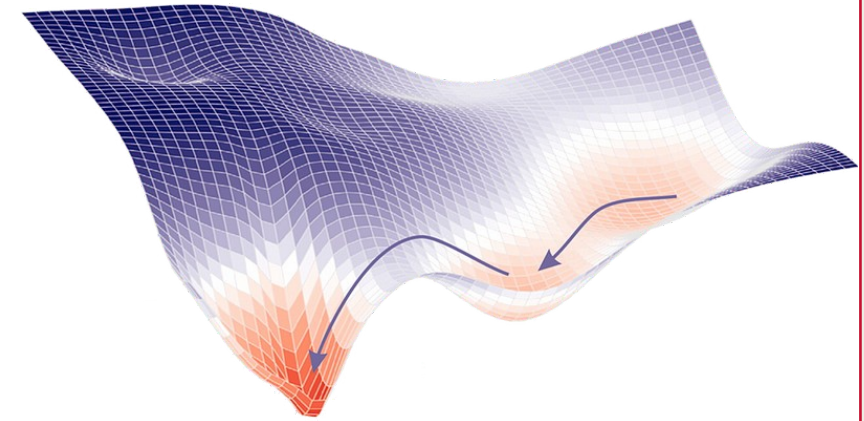
- ReLu

- Tanh

- Sigmoid

- Output Layer Activation

- Sigmoid





Python